Technical Report (Mar 2026)

Nickel-Zinc Ferrite Crack Behavior: ML-Based Study on Quasi-Rigid Polyurethane (PUR) Loading and Center Pin Diameter Effects

Mohamad Azim Alway [a,*], Mohd Ashraf Hamzah [b], Azlan Abu Asan@Hasan [b], Megat Iskandar [a]

[a] Department of Process Engineering, Murata Manufacturing Co. Ltd, Malaysia
[b] Department of Machine Engineering, Murata Manufacturing Co. Ltd, Malaysia

***Abstract*— Top-flange cracking in Ni–Zn ferrite inductors commonly originates from severe stress concentration induced by the flat-end cylindrical center pin during the wire-press process. The conventional 3.0 mm center pin acts as a localized indenter, amplifying tensile stresses around the ferrite top-flange perimeter and increasing the likelihood of brittle fracture. This study examines how press displacement, press speed, press time, and chuck-wear conditions govern fracture probability, while also characterizing how a polyurethane (PUR) loader stiffens when confined within a metal housing, transforming into a quasi-rigid element. Indentation tests using 3.0 mm and 3.8 mm pins were performed, followed by interpretable machine-learning analysis to quantify feature contributions and identify safe operating windows. Results show that enlarging the center pin diameter to 3.8 mm reduces peak stress concentration and lowers crack-risk occurrence by up to 58.33%, particularly within the critical displacement zone of 0.06-0.10 mm. These findings provide a practical and data-driven pathway for enhancing mechanical reliability in ferrite core inductor assembly.**

***Index Terms*—Ni-Zn Ferrite, PUR, Center Pin, Chuck Clamper, Wire Press Process, Stiffness, SHAP, PDP**

I. INTRODUCTION

Nickel-Zinc ferrite materials have long been used in modern electronic and automotive applications due to their functional properties, including high resistivity, high magnetic permeability, and excellent thermal stability. These characteristics make them dependable materials for inductors, ceramics capacitors, and similar components where electromagnetic reliability is important. Despite these advantages, Ni–Zn ferrites remain brittle ceramic materials with inherently low fracture toughness and tensile strength as reported by Yoshida et al [2], making them highly susceptible to crack initiation under localized mechanical loading.

In industrial manufacturing environments, Ni-Zn ferrite core is clamped inside a confined chuck component at the sides of top flange, where polyurethane (PUR) element acts as a quasi-rigid loader. During pressing process onto bottom flange area, a flat-end cylindrical center pin with 3.0 mm diameter will behave as an indenter pin by applying a highly concentrated load onto top flange of the ferrite core. The applied force is converted into localized stress at the contact interface, increasing the risk of stress concentration near center pin contact area. Prior studies [5],[7] have shown that flat-end cylindrical indenter pin (similar concept with center pin) can produce significant perimeter stress, peak localized loading, and bending stresses, which increase the likelihood of crack initiation in brittle materials such as Ni-Zn ferrite. There are 3 operating parameters that further influence the stress response during this process assembly, which is press displacement, press speed, and press time from polyurethane (PUR) interface. These mechanical interactions highlight how center pin geometry, loading parameters, and brittle characteristics contribute to manufacturing process assembly. To the authors' knowledge, previous studies have not combined indentation mechanics with interpretable machine-learning analysis to quantify crack-risk behavior in Ni–Zn ferrites.

II. THEORETICAL BACKGROUND

A. Material Stiffness Characterization

The wire press process plays a critical role in ensuring that the copper wire is properly seated within the electrode region of the Ni–Zn ferrite core during assembly. To facilitate this process, a polyurethane (PUR) element is commonly used as the loader material due to its ability to deform under load and its relatively low stiffness compared to a fully rigid metal plate. This deformability helps the PUR adapt to variations in the wire and ferrite geometry, enabling stable seating of the wire during compression.

However, the PUR is confined within a metal housing so that it can be integrated with machine components during wire-press process assembly. When PUR is compressed under confinement, its deformation is restricted, and this confinement causes the PUR to behave significantly stiffer than in its free, unconstrained state. As a result, the stiffness of PUR increases because the metal walls limit lateral expansion, a behavior like

*Corresponding author.
Email: mazim.alway@murata.com (Mohamad Azim, ME11253)

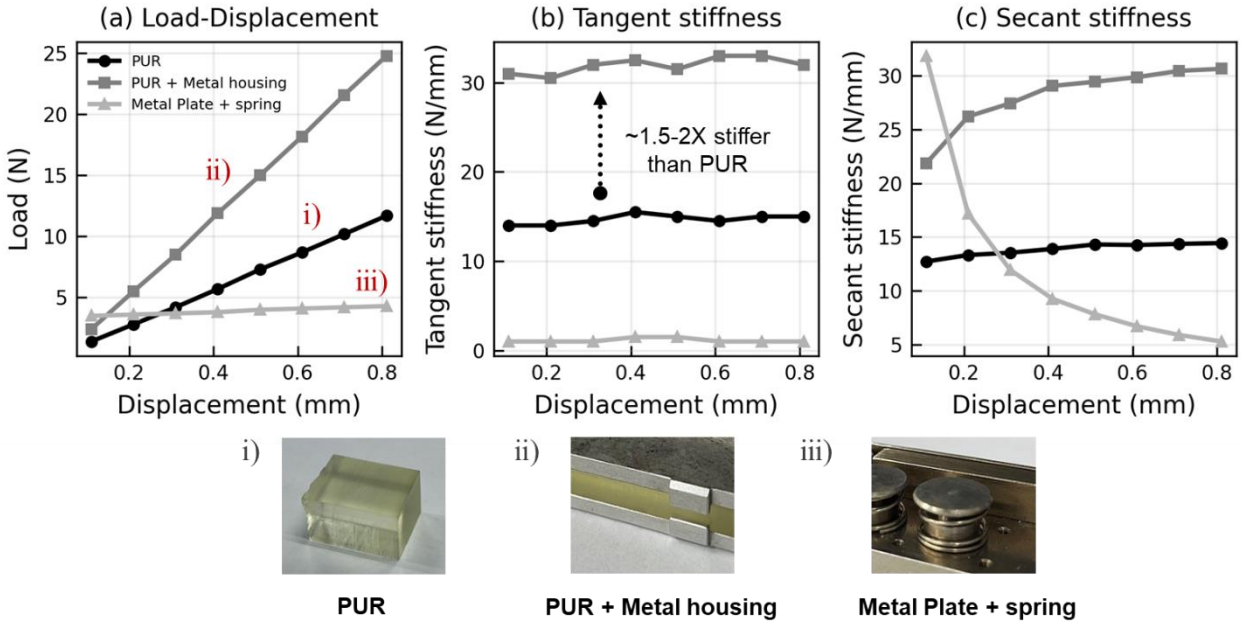


Fig. 1 a) Load vs. Displacement Response, **b)** Tangent Stiffness Curve, **c)** Secant Stiffness Curve

densification under compression. This confinement effect transforms the PUR from a soft, compliant material into a quasi-rigid loader during the pressing process, which directly influences the stress transfer onto the Ni–Zn ferrite core.

This behavior is illustrated in Fig. 1, where the Load–Displacement curve (Fig. 1a) shows that the PUR alone exhibits the lowest load response, while the PUR confined in a metal housing displays a much steeper slope, indicating higher effective stiffness. The tangent stiffness plot (Fig. 1b) further confirms this, showing that the PUR with metal housing condition is approximately 1.5–2 times stiffer than the free PUR. Similarly, the secant stiffness curve (Fig. 1c) highlights how the global stiffness of the PUR with housing configuration increases rapidly with displacement, approaching the behavior of a rigid plate-spring assembly. These observations confirm that the PUR with metal housing assembly behaves mechanically closer to a rigid spring element, a finding that is essential for interpreting the elevated stress transmission observed in the indentation experiments.

To sum up, when the PUR is constrained by metal walls, its lateral expansion is restricted, causing the material to stiffen sharply under compression. This confinement transforms the PUR from a compliant polymer into a quasi-rigid loading element during wire-press operation.

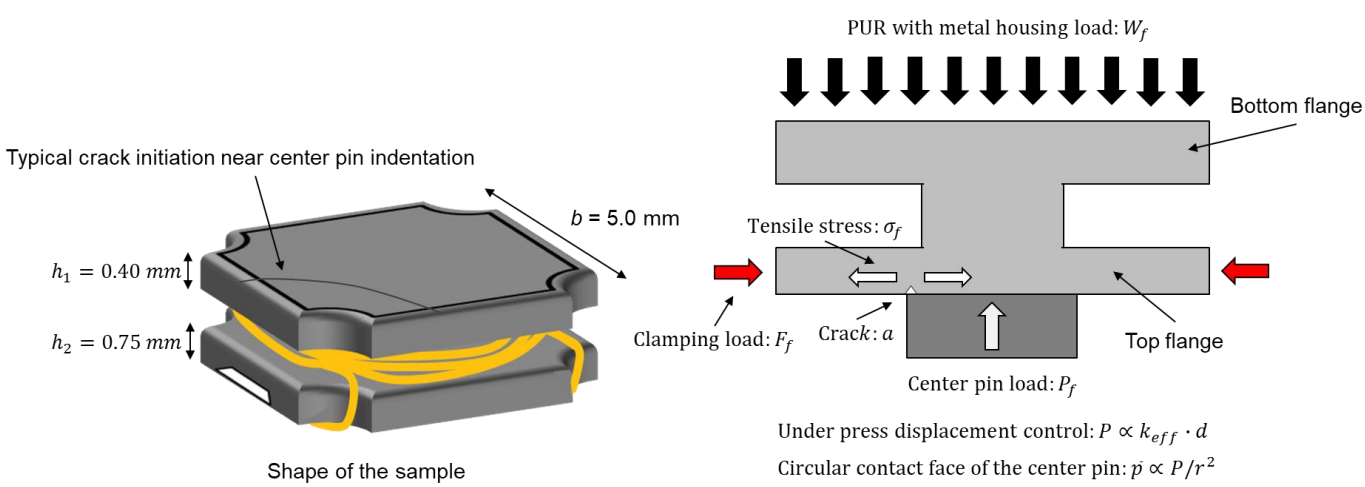


Fig. 2 Schematic Diagram of Localized Indentation (Flat-End Cylindrical Center Pin) on an Edge-Clamped Top Flange

B. Indentation Mechanics: Flat-end Cylindrical Center Pin

During wire press loading, the flat-end center pin behaves as a flat punch acting on the ferrite top flange. In displacement-controlled pressing, the reaction load increases proportionally to the effective stiffness of the loading stack, $P \propto k_{eff} \cdot d$. This is because the center pin's contact face is (nominally) a flat circle of radius r (half the pin's flat diameter), the average contact pressure grows as $\bar{p} \propto P/r^2$. Together, these relations explain the tensile hotspot shown in the schematic Fig. 2 as the imposed displacement d increases, a larger P is transmitted through the same contact area, intensifying σ_f beneath and around the punch and the stress then channels toward the side clamps region, where cracks can propagate once initiated near the punch perimeter. In practical terms, this means that even a small increase in applied displacement can produce a disproportionately large rise in indentation load when the loader stack becomes stiffer, such as under PUR-confinement conditions.

Crucially, the material stiffness characterization in the previous section showed that the PUR insert becomes quasi-rigid when confined in its metal housing, which elevates k_{eff} compared with an unconstrained PUR pad. That finding feeds directly into the indentation mechanics, where a higher k_{eff} means small increments in d generate disproportionately larger P at the same contact radius r, steepening the rise in $\bar{p}$ and intensifying stress localization at the top flange. In short, the stiffened loader measured earlier explains why the center pin indentation is highly sensitive to press displacement and why crack initiation commonly appears at the center pin perimeter region.

III. DESIGN OF EXPERIMENTAL

A. Experiment Setup

Ni–Zn ferrite components that fully met product dimensional specifications were selected as test samples. Each ferrite core was clamped inside a wear-conditioned chuck assembly, which constrained the top-flange edges to reproduce the boundary conditions of the actual wire-press station. A combined PUR and metal-housing fixture served as the loading interface. The PUR element, positioned inside the metal housing, transmitted compressive force while simultaneously experiencing confinement-induced stiffness elevation. A flat-end cylindrical center pin 3.0 mm and 3.8 mm in diameter, made direct contact with the ferrite top flange and acted as the controlled indentation source during all displacement-driven loading sequences.

All compression experiments were conducted using an Aikoh Engineering loading machine equipped with a calibrated 500 N load cell to ensure accurate force measurement. The machine applied displacement-controlled loading to the bottom flange region while the center pin simultaneously imposed localized indentation on the top flange. The setup enabled systematic variation of key process parameters, which included press displacement, press speed, press time, and center pin diameter. Each loading condition was executed under the same clamping constraints to ensure consistent mechanical boundaries, allowing direct comparison of how parameter variations influenced crack initiation. All test results including process variable parameters and crack failure outcomes were logged and subsequently used as inputs for machine-learning analysis.

B. Machine Learning Method

This study uses supervised classification to quantify how strongly manufacturing parameters influence crack failure in Ni–Zn ferrite cores during the wire-press operation. Four complementary model families were adopted, which is Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and XGBoost (XGB), to ensure the feature-importance conclusions are robust across different inductive biases (linear, margin-based, bagging, and boosting). The output variable is Crack Failure (binary: 1 = crack, 0 = no-crack). Predictors comprise routinely logged process parameters:

1) Press Displacement (mm)

2) Press Speed (mm/min)

3) Press Time (s)

4) Center Pin Diameter (mm)

The dataset (N=200) was split into Train and Validation (70%) and Test (30%) using stratified sampling to preserve the class ratio. Within 70%, a Validation set (30% of Train) was further held out for threshold tuning (overall proportions: 49% train,

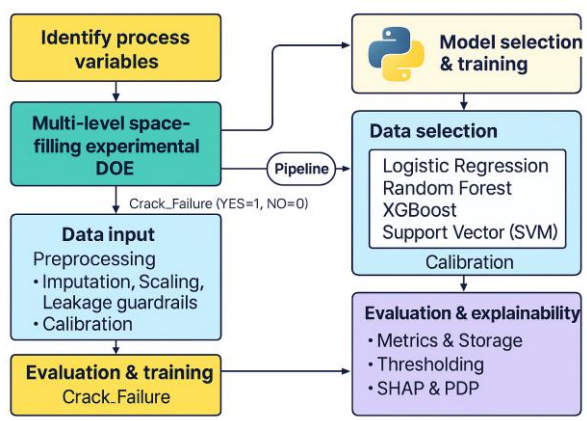


Fig. 3 Machine Learning Workflow for Crack Prediction

21% validation, 30% test). All experiments used a fixed random seed (RANDOM STATE = 42) for reproducibility. The focus of the study was not to compare model accuracy but to understand feature contributions. The model probabilities were analyzed using a unified Kernel SHAP framework. Kernel

SHAP was applied to the predicted probability of crack failure, using a consistent background sample across all models to ensure comparability. In other hand, partial dependence plots (1D) and a 2D risk map (Press Displacement × Pin Diameter) were used to visualize crack risk behavior and identify safer operating windows.

IV. RESULTS AND DISCUSSION

A. Effects of Feature Contributions on Crack Failure

Table. 1 indicates the performance of the four ML models on the held-out test dataset. Logistic Regression achieved the highest overall F1-score (0.9250) and ROC-AUC (0.9581),

Table. 1. Performance Comparison of ML Models on the Test Dataset

Model	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Logistic Regression	0.8810	0.9737	0.9250	0.9581	0.9743
Random Forest	0.9167	0.8684	0.8919	0.9420	0.9675
XGBoost	0.8649	0.8421	0.8533	0.9414	0.9663
SVM	1.0000	0.9259	0.8333	0.9318	0.9600

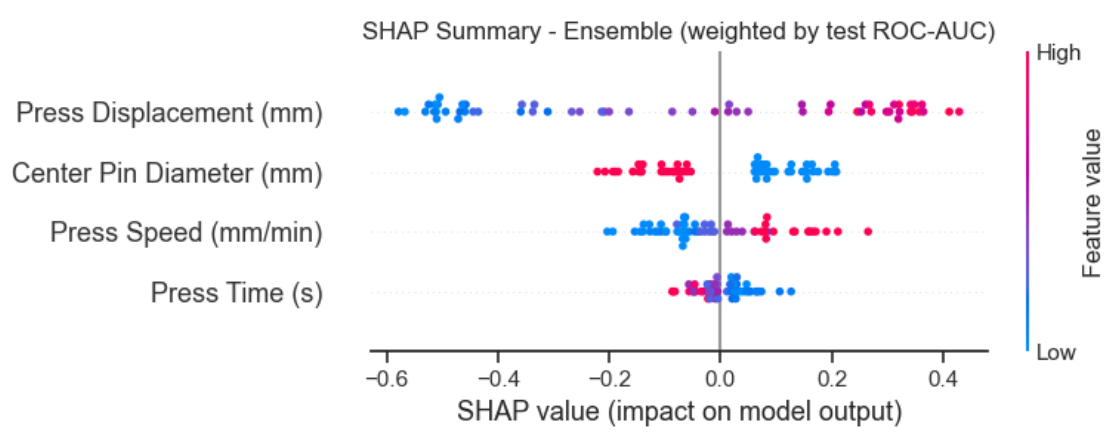
showing strong probability separability despite its linear nature. Random Forest and XGBoost both delivered competitive ROC-AUC values (0.9420 and 0.9414), reflecting stable nonlinear classification behavior, although their F1-scores were slightly lower due to class imbalance. The SVM model produced perfect precision (1.0000) but showed the lowest F1-score among all models because its recall was comparatively lower (0.9259), leading to an overconfident but less balanced decision boundary. Across all models, PR-AUC values exceeded 0.96, confirming that the predictors provided reliable probability estimates even under imbalanced crack/no-crack ratios. These results demonstrate that all four models are sufficiently accurate and well-calibrated to support the subsequent SHAP-based interpretability analysis. Since the objective of this study is to understand feature contributions rather than to optimize model accuracy, the comparable performance across model families ensures that the SHAP findings reported in Sections IV-A and IV-B are robust and not model-specific.

The ensemble SHAP summary plot in Fig. 4 a), weighted by each model's ROC-AUC performance, reinforces the trends observed in Table 1 and Fig. 4 a) by showing that press displacement exerts the strongest positive influence on predicted crack probability, with higher displacement values producing consistently positive SHAP contributions that drive the model toward crack-failure classification. Center pin diameter appears as the second most influential feature, where larger diameters yield predominantly negative SHAP values, indicating a protective effect that lowers predicted failure risk by reducing stress concentration at the top-flange interface. In contrast, press speed and press time cluster tightly around zero with comparatively small SHAP magnitudes and mixed color gradients, confirming their minimal contribution to classification outcomes across all four models. This ensemble SHAP distribution therefore aligns directly with the model-performance trends in Table 1, demonstrating that despite differences in learning mechanisms, all models consistently identify displacement as the critical driver of brittle fracture and pin diameter as the primary mitigating factor, while speed and time remain secondary parameters within the tested operating ranges.

Fig. 4 b) shows the unified feature-importance analysis generated using Kernel SHAP applied consistently across the four classification models (Logistic Regression, SVM, Random Forest, and XGBoost). Despite the different learning mechanisms of these models, interestingly all four exhibit a highly consistent pattern in which press displacement is shown as the dominant contributor to crack failure. This pattern aligns with the underlying mechanics of the process, where press displacement directly controls the applied load pathway and therefore dominates the formation of stress concentration at the top flange. As the displacement increases, the quasi-rigid PUR-metal housing transmits higher indentation force near flat-end cylindrical center pin region onto the ferrite core. The SHAP values therefore validate that displacement is the most critical explanatory process variable for predicting crack initiation under practical operating conditions.

The second most influential variable across the models is center pin diameter, with larger diameters consistently associated with a lower predicted probability of cracking. This observation is mechanically intuitive, suggests the changes of

a)



b)

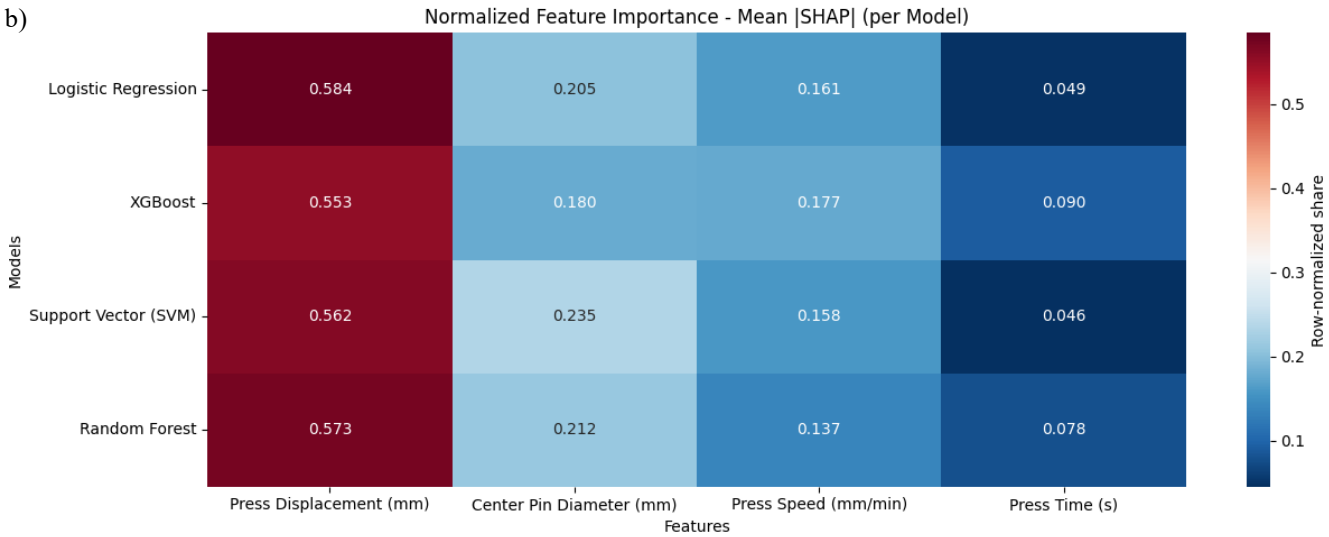


Fig. 4 a) Ensemble SHAP Summary Plot (Weighted by Test ROC-AUC), **b)** All Models Explained on Probability Scale via Kernel SHAP Heatmap; N=200

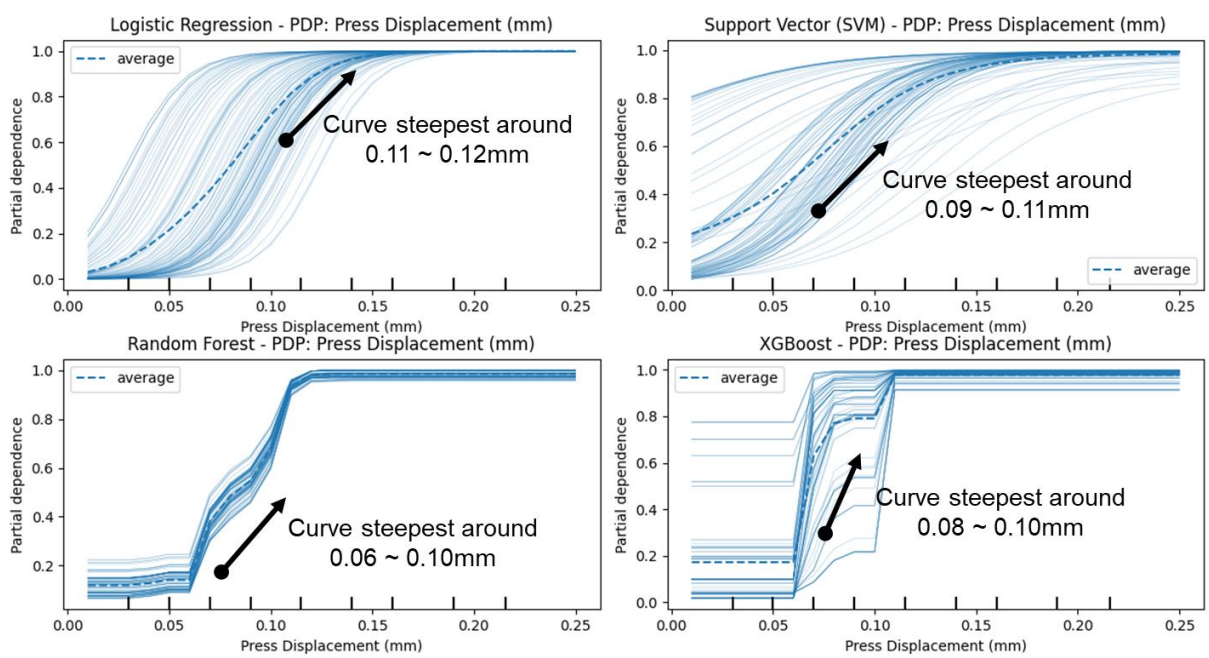
center pin diameter will affect the effective contact area, thereby changing the local stress intensity at the loading interface. The SHAP analysis strengthens this conclusion by showing that this benefit appears robust across linear, nonlinear, bagging, and boosting models, confirming that pin diameter directly influences how load is distributed into the brittle ferrite substrate.

However, press speed and press time show consistently lower importance across all models, indicating that within the tested operating ranges, these parameters do not substantially influence the stress state that leads to crack initiation. This suggests that, under the tested operating range, the cracking mechanism is governed more by peak stress concentration than by dynamic temporal effects. Press Speed and Press Time likely act as secondary modifiers of the mechanical response rather than primary initiators of structural failure. The consistency of this low-importance ranking across all four models indicates that, from a process-control perspective, optimizing press displacement and center pin diameter process variables substantially greater benefit for reducing crack risk than adjusting speed or time alone.

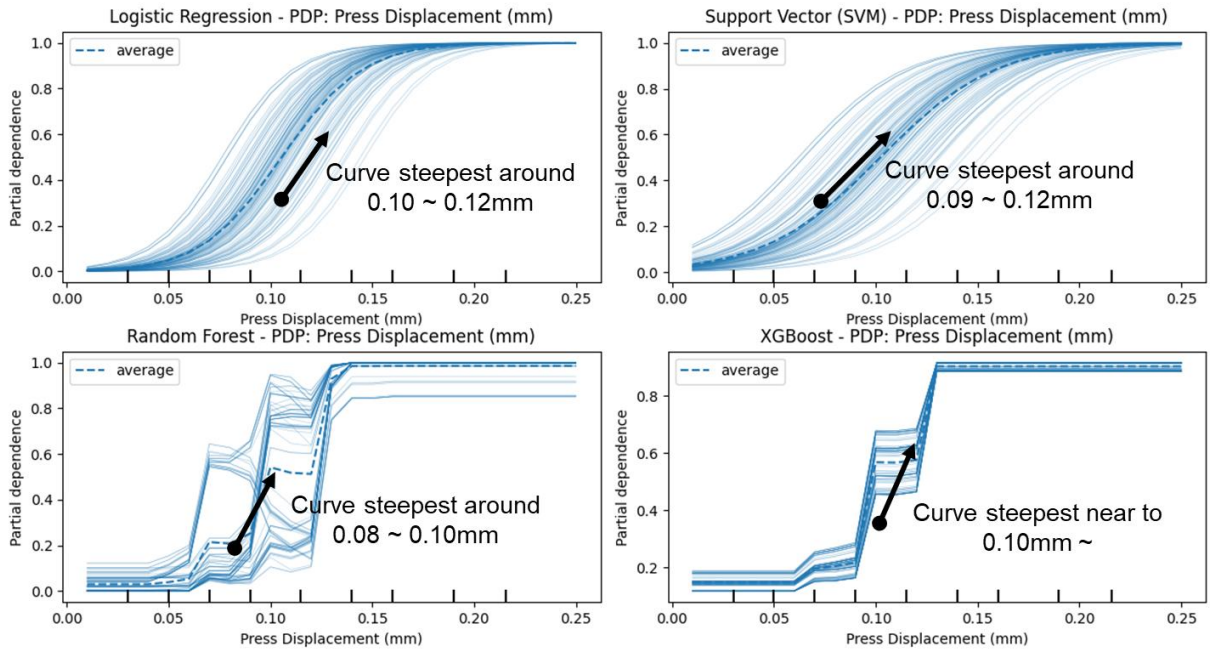
B. Effects of Center Pin Diameter

The diameter of the center pin significantly influences how mechanical stress is distributed across the top-flange region of the Ni–Zn ferrite core during the pressing operation. With a 3.0 mm flat-end cylindrical center pin, the smaller contact area between the pin and the ferrite surface concentrates the applied

a)



b)



c)

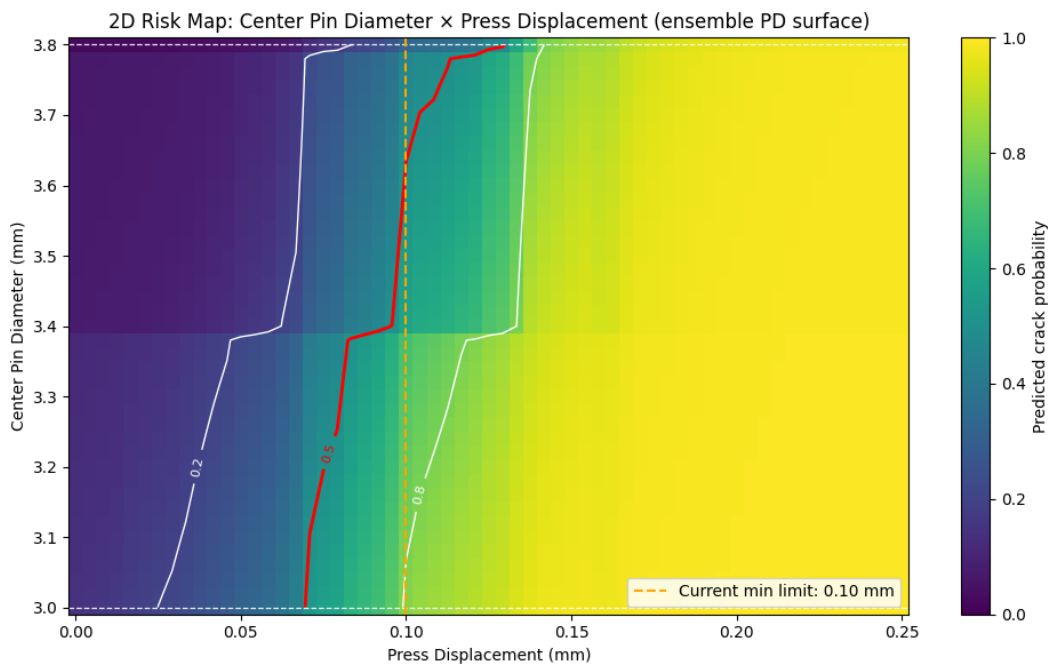


Fig. 5 a) Partial dependence of Press Displacement (mm) on a 3.0 mm Center Pin Diameter across Models, **b)** Partial dependence of Press Displacement (mm) on a 3.8 mm Center Pin Diameter across Models, **c)** 2D Crack-Risk Surface Across Center Pin Diameter and Press Displacement

load, thereby increasing the likelihood of crack initiation. As shown in Fig. 5 a), all models consistently indicate that with a 3.0 mm center pin diameter, once the press displacement control window enters the range of approximately 0.06–0.11 mm, the risk of cracking rises sharply before eventually reaching saturation. In contrast, Fig. 5 b) shows increasing the center pin diameter to 3.8 mm shifts the onset of high crack risk to press displacement levels above approximately 0.10 mm, indicating that the 3.8 mm pin distributes load over a larger contact area, effectively reducing local stress intensity transmitted into the brittle ferrite material. This geometric advantage directly lowers the peak stress gradient and slows the propagation of stress waves toward the chuck clamped boundaries. As a result, the 3.8 mm center pin diameter is safer and gives more tolerance to press displacement control window.

Fig. 5 c) shows a comparison between 3.0 mm and 3.8 mm pins, clearly indicates that the larger center pin consistently shifts the crack risk curve downward across the entire press

displacement range. By adopting the 3.8 mm center pin leads to reduces crack risk by up to 58.33%. This reduction is most evident in the critical press displacement "sensitive zone" where ferrite material behavior transitions toward brittle fracture. In this region, even small changes in load distribution significantly influence fracture probability, and the larger center pin diameter effectively mitigates this risk by preventing stress spikes associated with the smaller 3.0 mm pin. This shift in crack-risk behavior highlights the strong geometric sensitivity of the indentation interface and reinforces the mechanical insights drawn from the stiffness characterization.

In summary, increasing the center pin diameter not only reduces localized stress intensity but also broadens the safe operating window of the wire-press process. The 58.33% crack-risk reduction demonstrates that geometric optimization of the center pin is highly effective and practical engineering solution for improving ferrite mechanical reliability during inductor assembly.

V. CONCLUSIONS

With the aim of reducing top-flange cracking in Ni–Zn ferrite inductors during the wire-press process, the mechanical behavior under PUR-metal housing loading, the influence of center pin geometry, and the dominant process parameters were investigated through stiffness characterization, indentation tests, and interpretable machine-learning analysis. As a result, the following findings were obtained.

1) Indentation tests confirmed that the flat-end cylindrical center pin generates a strong tensile hotspot at the center pin perimeter. Cracking consistently initiated at this region due to localized stress concentration magnified by the elevated stiffness of the PUR with metal housing assembly.
2) Machine-learning interpretation using Kernel SHAP revealed that press displacement is the most influential parameter governing crack occurrence, followed by center pin diameter. Press speed and press time contributed minimally within the tested operating window, indicating that fracture is governed primarily by peak stress concentration rather than rate-dependent effects.
3) Increasing the center pin diameter from 3.0 mm to 3.8 mm broadened the effective contact area, reduced stress intensity, and shifted the onset of high-risk cracking to larger displacement values.

ACKNOWLEDGMENT

The authors would like to express their sincere appreciation to the Process Engineering and Machine Engineering Departments at Murata Manufacturing Co., Ltd., Malaysia, for providing the experimental facilities, technical guidance, and support required to complete this study. The authors also acknowledge the contributions of the machine engineering team for assisting with components fabrication, equipment calibration, and data acquisition during the evaluations. Their cooperation and expertise were essential in carrying out the mechanical tests and analysis presented in this work.

DATA AVAILABILITY

Data will be made available on request.

REFERENCES

[1] J. Hwang, M. Choi, H.-S. Shin, B.-K. Ju, and M. Chun, "Structural and magnetic properties of NiZn ferrite nanoparticles synthesized by a thermal decomposition method," Applied Sciences, vol. 10, no. 18, art. no. 6279, 2020, doi: 10.3390/app10186279.

[2] H. Yoshida, T. Hiratani, and T. Tagawa, "MnZn ferrite with higher strength," JFE Technical Report, no. 27, pp. 28–34, Mar. 2022.

[3] A. C. Hee, M. Mehrali, H. S. C. Metselaar, M. Mehrali, and N. A. A. Osman, "Comparison of nanostructured nickel zinc ferrite and magnesium copper zinc ferrite prepared by water-in-oil microemulsion," Electronic Materials Letters, vol. 8, no. 6, pp. 639–642, 2012, doi: 10.1007/s13391-012-2029-7.

[4] M. Shariyat, "Novel finite-thickness contact models for composite sandwich plates with auxetic viscoelastic cores under low-velocity impact: effects of indenter shape and boundary conditions," Composite Structures, vol. 372, 119596, 2025, doi: 10.1016/j.compstruct.2025.119596.

[5] A.-H. S. S. Said, A. Seif, E. Ghandourah, W. Nureldeen, and I. M. R. Najjar, "Balancing impact and indentation performance of hybrid GFRE–steel mesh laminates under various indenter shapes and loading scenarios," Journal of Materials Research and Technology, vol. 38, pp. 4860–4880, 2025, doi: 10.1016/j.jmrt.2025.08.275.

[6] S. I. Okocha, P. Y. B. Jar, and M. T. Hendry, "Measurement of fracture toughness in high-strength alloys via modified limit load analysis using flat-end cylindrical indenter," Theoretical and Applied Fracture Mechanics, vol. 134, 104740, 2024, doi: 10.1016/j.tafmec.2024.104740.

[7] L. Xu, L. Kong, H. Zhao, S. Wang, S. Liu, and L. Qian, "Mechanical behavior of undoped n-type GaAs under the indentation of Berkovich and flat-tip indenters," Materials, vol. 12, no. 7, 1192, 2019, doi: 10.3390/ma12071192.

[8] A. Ochoa Brezmes and C. Breitkopf, "Influence of indenter tip diameter and film thickness on flat indentations into elastic–plastic films by means of the finite element method," Thin Solid Films, vol. 653, pp. 49–56, 2018, doi: 10.1016/j.tsf.2018.03.016.

[9] Y. Xiao, J. Yin, X. Zhang, X. An, Y. Xiong, and Y. Sun, "Mechanical performance and cushioning energy absorption characteristics of rigid polyurethane foam at low and high strain rates," Polymer Testing, vol. 109, 107531, 2022.

[10] W. Eccles, I. Sherrington, and R. D. Arnell, "Frictional changes during repeated tightening of zinc plated threaded fasteners," Tribology International, vol. 43, pp. 700–707, 2010, doi: 10.1016/j.triboint.2009.10.010.

[11] Chen, Tianqi, and Carlos Guestrin. "XGBoost: A Scalable Tree Boosting System." arXiv preprint arXiv:1603.02754 (2016).

[12] H. Agarwal, M. N. Shariff, and S. Mangalathu, "Prediction of flexural crack width of reinforced concrete beams using interpretable machine learning algorithms," Construction and Building Materials, vol. 485, 141628, 2025, doi: 10.1016/j.conbuildmat.2025.141628.